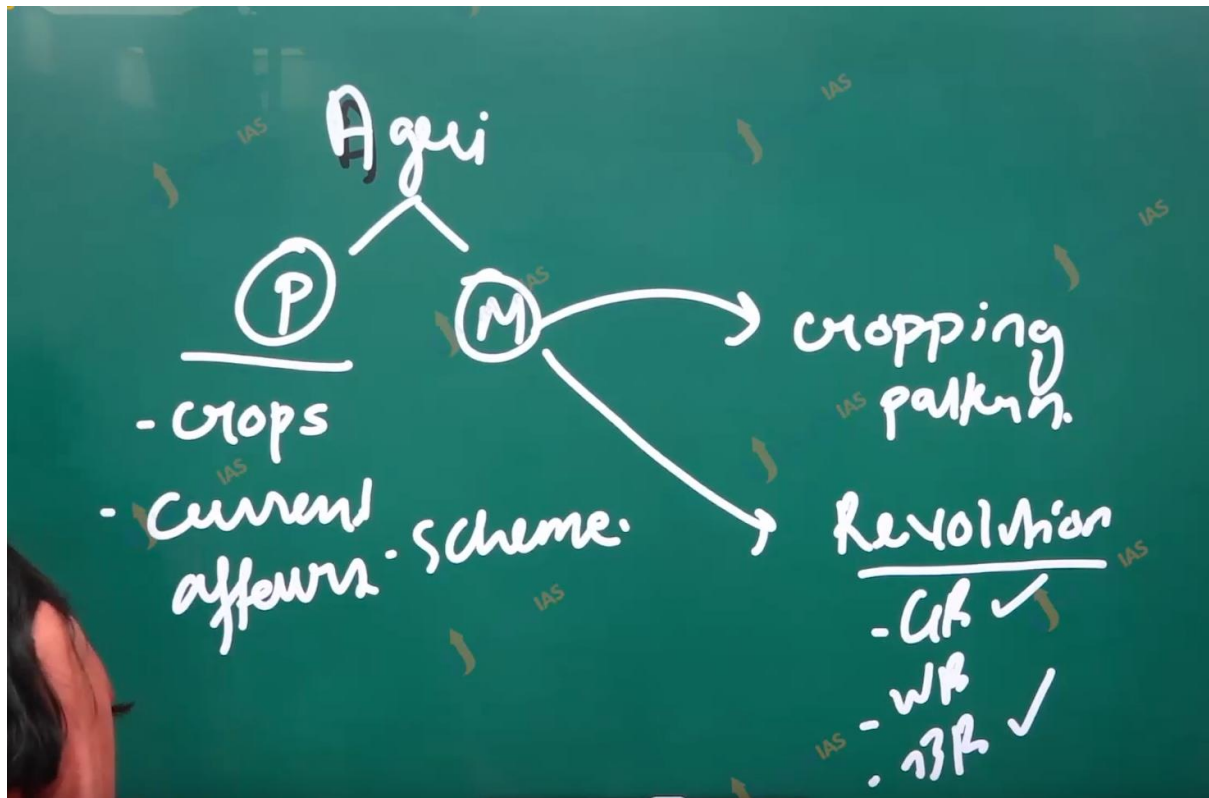


(39) Agriculture



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Major Crop: Rice

- It is a kharif crop which requires high temperature, (**above 25°C**) and high humidity with annual rainfall above **100 cm**.
- In the areas of less rainfall, it grows with the help of irrigation such as **Punjab, Haryana and Western Uttar Pradesh and parts of Andhra Pradesh**.
- It is grown in the plains of north India, coastal areas and the deltaic regions.
- Soil, Alluvial Soil



INDIA RICE

MAJOR AREA

MINOR AREA

0 200 400 600 km

LevelUp IAS

Agriculture:

Major crops:

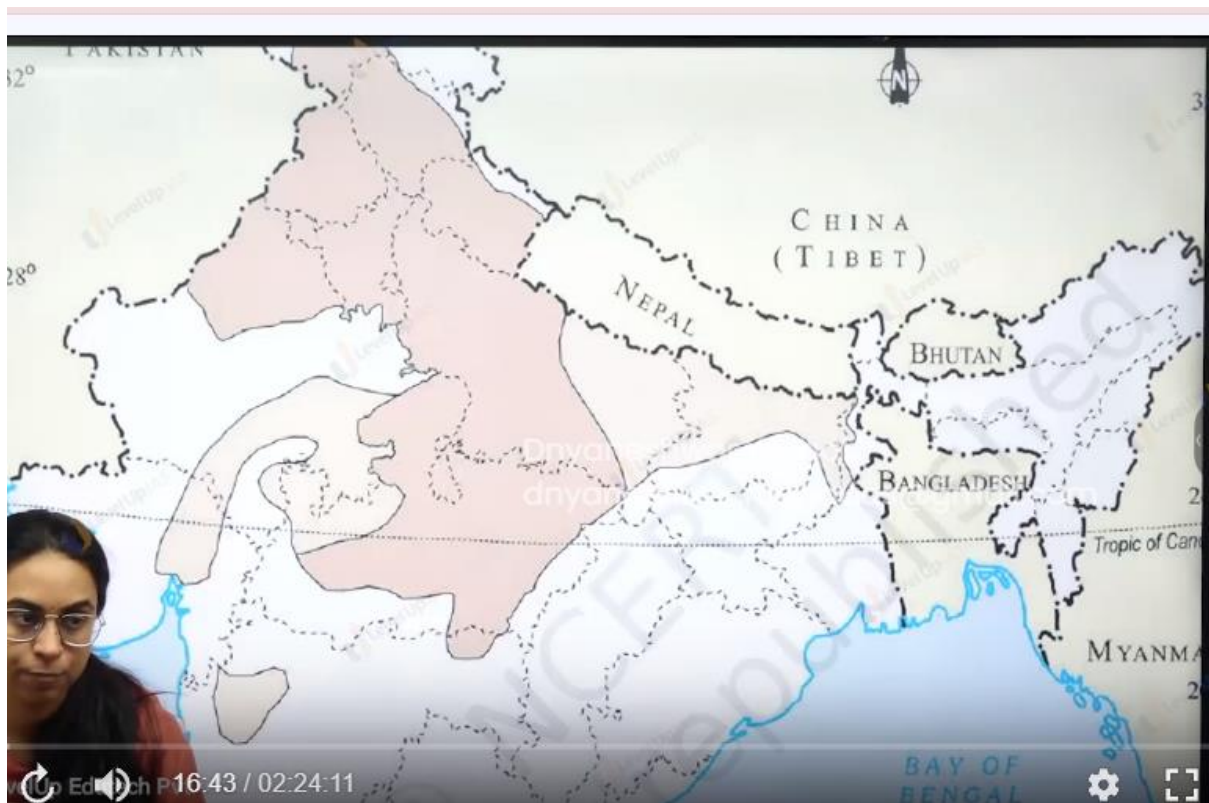
1. Rice:

- a. It is a kharif crop. It required a high temperature (greater 25 degree Celsius) and high humidity rainfall of 100 cm.
 - b. In the area of leases rainfall, it can grow with the help of irrigation. In the area Panjab Haryana western UP.
 - c. Rice is majorly grown in the northeast India the northern plane. Coastal region and deltaic region.
 - d. Soils: alluvial soil, black cotton coils, loam soil (sand 40 silt 40 and clay 20).
 - e.
-



Wheat:

1. Second most important crop. Mainly grown northern and northern western Indian. It required a cool growing season and bright sunshine at a time of ripening.
2. It requires the rainfall of 50 75 cm. which is evenly distributed.
3. Two important zones where wheat is cultivated is Sat lag and Guanga plane and black cotton soil region of deccan. And therefore major wheat producing stage are Panjab Haryana UP MP Bihar rajasthan.



Tea:

1. It is beverages crop and grows on the plantation. It requires the cool climate and well distributed high rainfall 150-200 cm throughout the year.
2. Well distributed rainfall can help in the growth in tender leaves.
3. It can grow well on a gentle slope. Hilly regions it can grow in tropics as well as sub-tropics.
4. The flavor of the tea is imparted due to humidity, dew, fog, etc.
- 5.

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Major Crop: Tea

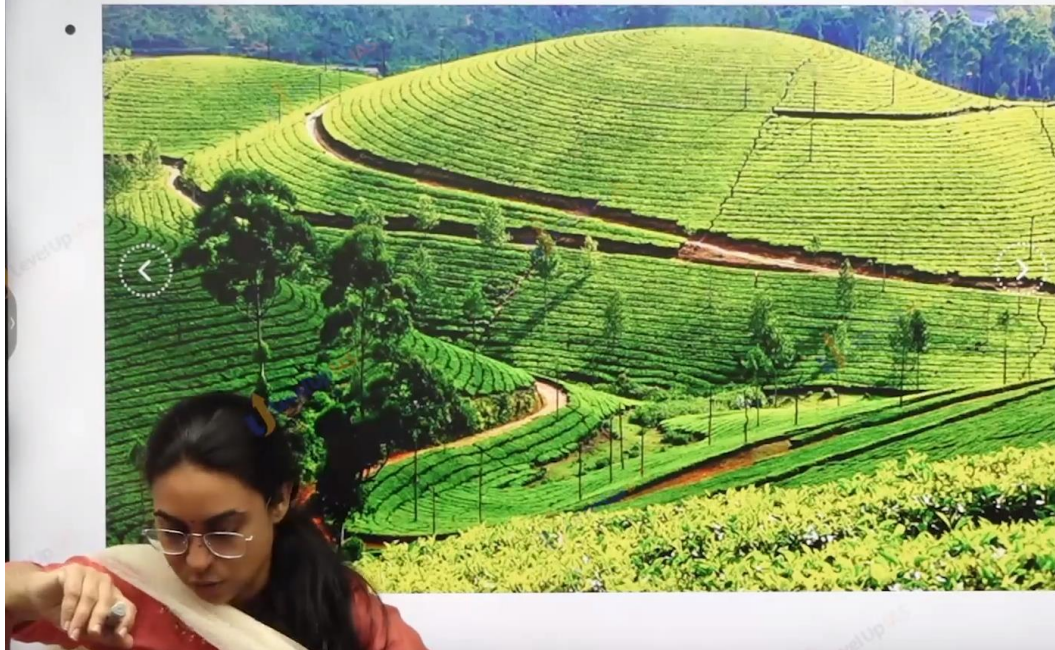
- Tea is a beverage crop grown on plantations.
- This requires cool climate and well distributed high rainfall (150-200 cm) throughout the year for the growth of its tender leaves.
- It needs well-drained loamy soils and gentle slopes.
- It can grow in tropical and sub-tropical climates.
- Tea bushes require frost-free temperature all through the year.
- It requires high humidity, heavy dew and morning frosts.
- It is a perennial crop.

Major Tea-Producing States In India



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Tea Estate



Coffee:

1. Coffee required warm and humid climate with well rain lomi soil.
2. Therefore hill slope are suitable for this crop.
3. Initially this crop is cultivated in baba budan hill kin Karnataka. Even till date 70% is confine to Karnataka.
4. Coffee cultivation is confine to nil Giri hill in Kerala , kar nakta, tamilnadu.
- 5.

Major Crop: Coffee

- Plantation found throughout Brazil; during the colonial period (16th - 18th centuries)



Dnyaneshwar mogal
dnyaneshwar.mogal.ias@gmail.com

Major Crop: Coffee

- Coffee requires warm and humid climate and well- drained loamy soil.
- Hill slopes are more suitable for growth of this crop.
- Low temperature or frost condition is harmful
- Excessive heat is also harmful
- Initially its cultivation was introduced in Baba Budan Hills
- Now its cultivation is confined to the Western Ghats in Karnataka, Kerala and Tamil Nadu



Jute:

1. It is also known as golden fiber. It grows well on alluvial soil it requires high temperature (25) heavy rainfall (125-200 cm) and humid climate.
 2. Jute is major cultivated along in west Bengal Andhra Pradesh Odisha and Bihar, Assam.
 3. It grows in tropical areas and in well drain fertile are where soil is renew every year.
 4. It is a kharif crop.
-

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Major Crop: Jute

- Jute was also known as the 'Golden Fibre'.
- It grows well on alluvial soil and requires high temperature, heavy rainfall (125-200 cm) and humid climate.
- This crop is grown in the tropical areas.
- Jute grows well on well-drained sandy or clayey loam

Major Jute-Production States In India



A woman is pointing at the map.

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Process of ratting:

1. It refers to the process of extracting jute fiber from the jute stem.
 2. Jute stem is kept under free-flowing water so have to facilitate easy removal of jute fiber.
 3. It is labor intensive process.
 - 4.
-



Cotton:

1. It requires the high temperature. Light rainfall (50-75) 210 frost free days. Bright sunshine for ripening and busting the crop.
2. It grows best on black cotton soil, and alluvial soil. It is a kharif crop and can be grown in tropical and subtropical area.
3. The moist weather during boll opening and harvesting is harmful as it leads to the spread of disease.
4. Can be grown in tropical and subtropical area. Ukraine is the northern limit of cotton.

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Major Crop: Cotton

- Cotton requires high temperature, light rainfall (50-75 cm, 210 frost-free days and bright sunshine for ripening and bursting crops.
- It grows best on black cotton soil of Deccan Plateau and alluvial soils.
- It is a kharif crop and requires 6 to 8 months to mature.
- Cotton is grown in tropical and subtropical areas (21-30°C).
- Maximum temperature during the blooming and opening of cotton is 30-35°C, promoting the growth of the crop.
- The northernmost limit of cotton is in the Himalayas.

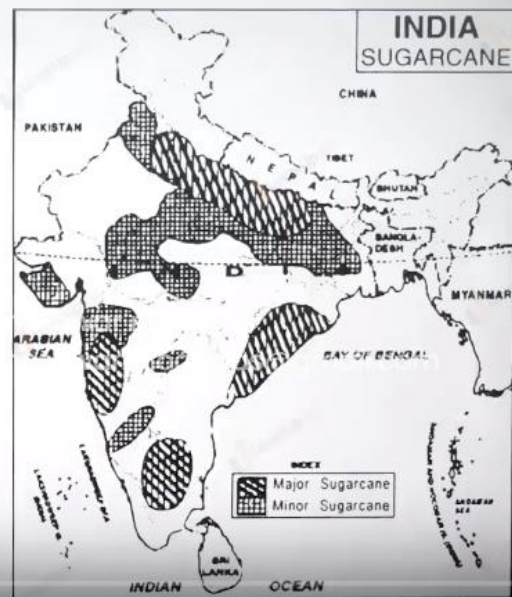
Major Cotton Growing Areas of India

Sugarcane:

1. It is plantation crop it requires the long rainy season of 8 month for it well developed growth.
2. Full grown sugar cane 8 to 12 to 15 months.
3. The conditions need to sugarcane are annually 75 to 100 cm temperature in the range of 21 to 27 degree.
4. It can be grown in the areas of low rainfall. It the help of irrigation support it can be grown in tropical as well as subtropical regions.
5. The soil need Is rich loamy soil. Which is neither too acidic nor alkaline.
6. The traditional belt of sugarcane is UP, Bihar. Later the crops are cultivated in the regions in Maharashtra (black cotton soil) Cauvery basin), Krishna, Godavari delta, and towards Panjab regions.
7. The greater sucrose yield in topical region.
- 8.
9. Ratooning : ratooning is pa
- 10.

Major Crop: Sugarcane

- It is a tropical as well as a subtropical crop.
- It grows well in hot and humid climate with a temperature of 21°C to 27°C and an annual rainfall between 75cm. and 100cm.
- Long rainy season of 8 months
- Irrigation is required in the regions of low rainfall.
- Temp above 40 and below 15 is not suitable
- Soil deep, rich loamy. Neither too



50:47 / 02:24:11



Sugarcane: Ratooning

- Method of cane propagation and harvesting in which a portion of stalk is left underground to give rise to a succeeding growth of cane (called the ratoon)
- Economical crops can be taken from one original planting.
- The yield of ratoon crops decreases each cycle, and at the end of the economical cycle all stumps are cut out and the field is replanted

RATOON CROP CYCLE



Ratooning:

1. The ratooning is practice of sugarcane proportions and harvesting in which a portion of stem is left which facilitates the faster growth of cane crop, normally sugarcane crop cannot be uprooted as it will take longer time for the growth of sugarcane. Therefore, ratooning is used.
2. The yield in the ratooning crops (seceding sugarcane crops) decreases each cycle. Therefore, at the end of the last economic cycle all stems and crops and ploughed out and the field is replanted.

LevelUp IAS
PYQ: 2011
To exit full screen, press Esc

A state in India has the following characteristics:

1. Its northern part is arid and semi-arid.
2. Its central part produces cotton.
3. Cultivation of cash crops is predominant over food crops.

Which one of the following states has all of the above characteristics?

(a) Andhra Pradesh
(b) Gujarat
(c) Karnataka
(d) Tamil Nadu

Handwritten notes:
a) Andhra Pradesh
b) Gujarat
c) Karnataka
d) TN

Ans: Gujarat.

AP has not black soil.

Gujarat has black soil.

Tamil Nadu do not produce the cotton.

And Karnataka not central part produce cotton but its upper part may be.

Arid meaning. Very dry and little rain.

No arid in Karnataka.

2011

Q) Salinization occurs when the irrigation water accumulated in the soil evaporates, leaving behind salts and minerals. What are the effects of Salinization on the irrigated land?

- a) It greatly increases the crop production
- b) It makes some soils impermeable
- c) It raises the water table
- d) It fills the air spaces in the soil with water

Ans is B

Does not increase production.

Impermeable means not allowing to fluid pass through.

2011

The lower Gangetic plain is characterized by humid climate with high temperature throughout the year. Which one among the following pairs of crops is most suitable for this region?

- a) Paddy and cotton.
- b) Wheat and jute.
- c) Paddy and jute.
- d) Wheat and cotton.

Ans is C

Meaning up bihar and orisa banga.

Weat and cotton required less water.

So paddy and jute.

Levelup IAS
PYQ: 2011

Among the following States, which one has the most suitable climatic conditions for the cultivation of a large variety of orchids with minimum cost of production, and can develop an export oriented industry in this field?

- (a) Andhra Pradesh
- (b) Arunachal Pradesh
- (c) Madhya Pradesh
- (d) Uttar Pradesh



Ans: Arunachal Pradesh.

Levelup IAS
Orchids: Plantation Crop: Horticulture

- 1. 70- 80% humidity,
- 2. Day temperature of 21 – 29 degree C and
- 3. Night temperature of 18 to 21 degree C is ideal for growing this tropical orchid.



PYQ: 2012

Which of the following is the chief characteristic of 'mixed farming'?

- (a) Cultivation of both cash crops and food crops
- (b) Cultivation of two or more crops in the same field
- (c) Rearing of animals and cultivation of crops together
- (d) None of the above

Ans : c

Cultivation of Two and more crops are mixed cropping ..

When you added the animals in to it it mixed farming.

PYQ: 2013

Consider the following crops:

1. Cotton
2. Groundnut
3. Rice
4. Wheat

Which of these are Kharif crops?

- (a) 1 and 4
- (b) 2 and 3 only
- (c) 1 and 3
- (d) 2 and 4

Kharif crops : rice required more water so in monsoon season.

And cotton. You are one of cotton producing farmer. You must know.

Karif : cotton and rice.

Ground nut.

LevelUp IAS
2014

Consider the following pairs:
Region well-known for the production of

1. Kinnaur : Arecanut
2. Mewat : Mango
3. Coromondal : Soyabean

Which of the above pairs is/are correctly matched?

(a) 1 and 2 only (b) 3 only
(c) 1, 2 and 3 (d) None of the above

01:20:49 / 02:24:11

LevelUp IAS
2014

Consider the following pairs:
Region well-known for the production of

1. Kinnaur : Arecanut
2. Mewat : Mango
3. Coromondal : Soyabean

Which of the above pairs is/are correctly matched?

(a) 1 and 2 only (b) 3 only
(c) 1, 2 and 3 (d) None of the above

01:21:51 / 02:24:11

Ans is : d None of above.

Mewat is in haryan, crime activity. Anti social element.

Areca nut : supari,

LevelUp IAS
2020

"The crop is subtropical in nature. A hard frost is injurious to it. It requires at least 210 frost-free days and 50 to 100 centimeters of rainfall for its growth. A light well-drained soil capable of retaining moisture is ideally suited for the cultivation of the crop." Which one of the following is that crop ? ☐

- (a) Cotton
- (b) Jute
- (c) Paddy
- (d) Sugarcane



Ans is : cotton A.

LevelUp IAS
2021

Among the following, which one is the least water-efficient crop?

- (a) Sugarcane
- (b) Sunflower
- (c) Pearl millet
- (d) Red gram



LevelUp IAS
Millets

1. Imp millet are: Jowar, bajra and ragi
2. Coarse Grain with very high nutritional value.
3. Ragi is very rich in iron, calcium, micro nutrients and roughage.

Jowar is the third most important food crop with respect to area and production.

1. It is a rain-fed crop mostly grown in the moist areas which hardly needs irrigation.
2. Major Jowar producing States are Maharashtra, Karnataka, Andhra Pradesh and Madhya Pradesh.

Bajra

1. grows well on sandy soils and shallow black soil. Major Bajra producing
2. Major Bajra producing states are Rajasthan, Uttar Pradesh, Maharashtra, Gujarat and Haryana.

Ragi

1. is a crop of dry regions and grows well on red, black, sandy, loamy and shallow black soils.
2. Major ragi producing states: Karnataka, Tamil Nadu, Himachal Pradesh, Uttarakhand, Sikkim, Jharkhand and Arunachal Pradesh.

INTERNATIONAL YEAR OF
MILLETS
2023



Millets:

1. It is grown in dry regions. The important millets are jowar bajra ragi.
2. They are called coarse grain with very high nutritional value.

3. They are low glycemic index meaning sugar in the blood. Diabetes
 4. They are recommended to diabetic people.
 5. Jawar is the 3 most important food crop with respect of area and production.
 6. It is a rain fed crop and grows in the moist area. which hardly need irrigation.
 7. The area where the jawar is grown are MP, MAHARASHTRA, KARNATAKA, Andra Pradesh.
 8. Bajra: bajra grows well on the sandi soil and sallow black soil.
 9. Major bajra producing states are Rajasthan and UP, Maharashtra. Gujarat, Telangana.
-

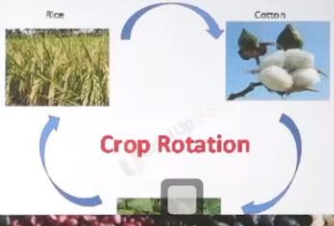
Ragi:

1. It grows on black soil sandy soil. Loamy soil, red soil.
 2. Ragi is majorly grown in Karnataka and tamandu and the hilly state of Uttarakhand Sikkim Jharkhand, Arunachal Pradesh, Himachal Pradesh.
-

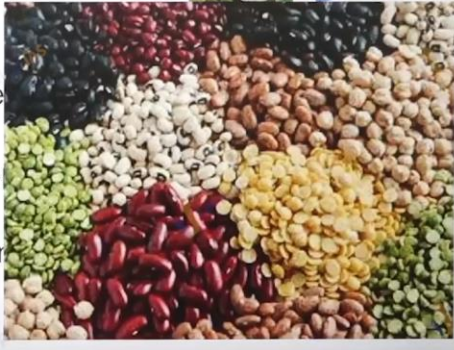
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Pulse

- India is the largest producer, consumer and importer of pulse.
- Major pulses grown in India are tur (arhar), urad, moong, masur, peas and gram.
- Pulses need less moisture and survive even in dry conditions.
- Being leguminous crops, all these crops except arhar help in restoring soil fertility by fixing nitrogen from the air.
- Pulses are mostly grown in rotation with other crops.
- Major pulse producing states
- Major growing areas: Madhya Pradesh, Rajasthan, Maharashtra, Uttar Pradesh and Karnataka.



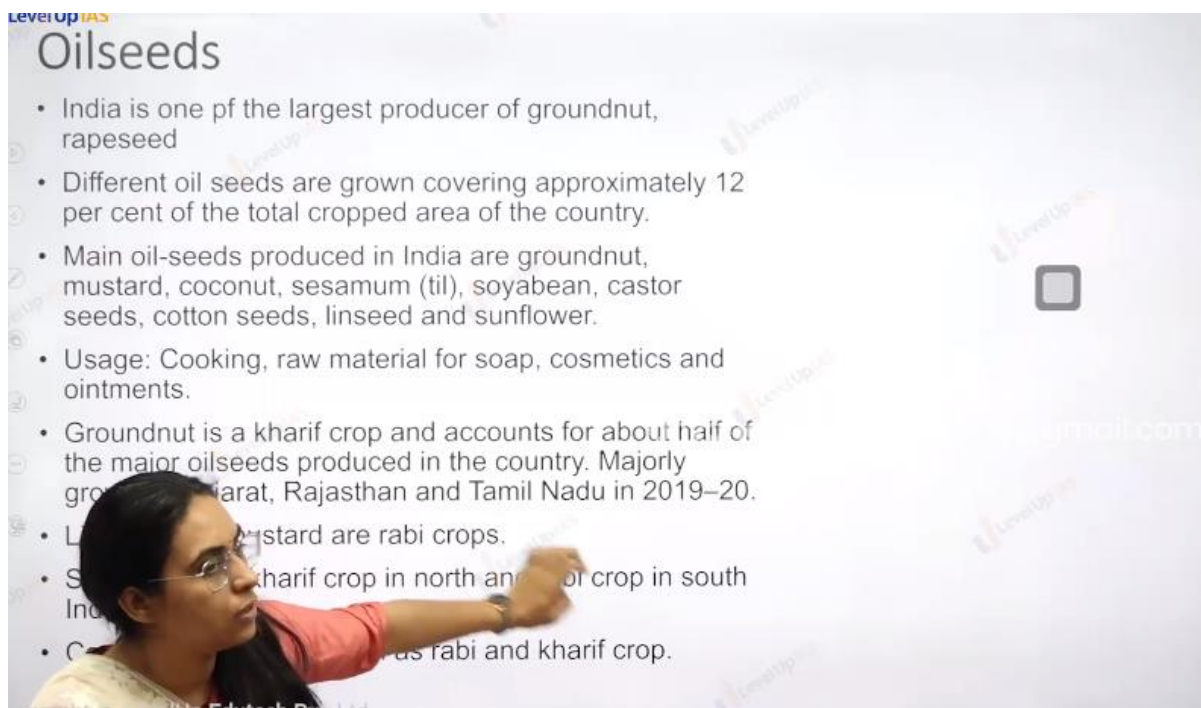
The diagram illustrates crop rotation with two crops: Rice and Cotton. Arrows show a cycle where Rice is followed by Cotton, and Cotton is followed by Rice. The text 'Crop Rotation' is written in the center of the cycle.



A large pile of various pulses, including red kidney beans, yellow lentils, and green lentils, is shown.

Pulses:

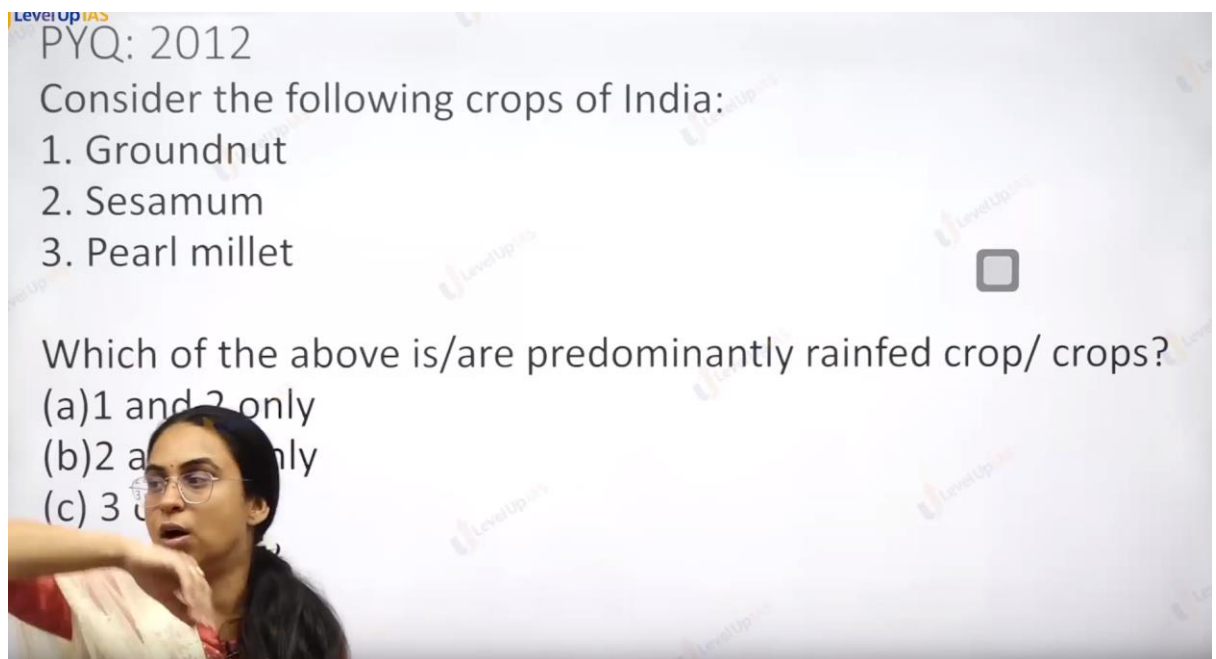
1. India is largest producer consumer and importer of pulses.
2. The pulses grown in India are pigeon pea, Tur (ahar, urad, moong , masur, peas ,gram.
3. Pulses can be grown in dry condition. Pulse is in form of crop rotation as they help in restoring soil fertility by fixing the nitrogen from the air, areas where pulse production dominate MP, Rajasthan, Maharashtra, UP, Karnataka.
4. Black gram can grow as kharif and rabi crop.
5. Gram is predominant plus accounting for 40% of total pulse production followed by tur 20% followed for urad 8% moong 8 %
6. Pulsed account for 20% area for crop production in India.
7. Rabi pulse account for 60% of total production.



Oil seeds:

1. India one of largest producer of ground nut and rapeseed.

2. Oil seeds cover the 12% of total crop area.
 3. Major oil seed produce are ground nut mustard, coconut, sesamum(til) , soyabean, castor seed, cotton seed linseed and sun flower.
 4. Oil seeds are used in cooking and raw materials for many industries like soap, cosmetics ointment.
 5. Ground nut accounts for half of major oil seed produce in the country.
 6. Some oil seeds are kharif crop and others are rabi cope. Ex. Cotton Is kharif crop whereas linseed and mustard are grown as rabi crop.
 7. Some of them are grown as kharif crop and rabi crop. Ex. Seasimum and caster seeds.
-



Ans: d all of above. Millet does not need more water.

Til no need to have separate irrigation.

PYQ: 2012

Consider the following crops of India:

1. Cowpea
2. Green gram
3. Pigeon pea

Which of the above is/are used as pulse, fodder and green manure?

- (a) 1 only
- (b) 2 only
- (c) 1 & 2 only
- (d) All of the above

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Ans is all of the above.

2019

Among the following, which one is the largest exporter of rice in the world in the last five years?

- (a) China
- (b) India
- (c) Myanmar
- (d) Thailand

Ans is India.

2020

With reference to pulse production in India, consider the following statements:

1. Black gram can be cultivated as both kharif and rabi crops.
2. Green-gram alone accounts for nearly half of pulse production.
3. In the last three decades, while the production of kharif pulses has increased, the production of rabi pulses has decreased.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) 2 and 3 only
- (d) 1 and 3 only

Ans:

1 only A.

2021

Among the following, which one is the least water-efficient crop?

- (a) Sugarcane
- (b) Sunflower
- (c) Pearl millet
- (d) Red gram
